AI ETHICS

AI for Whom?

By Danya Glabau, Industry Assistant Professor, Director of Science and Technology Studies, NYU Tandon School of Engineering

This module asks and answers the question, Who is artificial intelligence for? Using examples from existing uses of AI in healthcare and workplaces, it presents evidence that AI systems do not always help their intended users and constituencies. The module ends with a brief discussion of how to mobilize the presented case studies to improve AI systems evaluation, public consultation, and implementation in the public sector.

Lecture Transcript

0:04 Hello and welcome to the talk, "AI for Whom?" I am Danya Glabau, a professor at NYU Tandon School of Engineering. And we're going to go through this short lecture today to talk a little bit about who AI benefits, who it might harm and some of the implications for policy makers and people working in the public interest. So I'm going to start with just a little bit about me before we jump into the examples, and some ideas about what to do about them. So again, my name is Danya Glabau. I am in Industry Assistant Professor at NYU Tandon School of Engineering in the Technology, Culture and Society Department. I also direct the Science and Technology Studies major and minor, the second minor called Feminism and STEM. It's an interdisciplinary social science program that complements students' engineering education to introduce them to ideas about ethics and the social dynamics of science and technology. And in the major students specialize in both the social science perspective as well as technical engineering skills. My training started at Cornell University where I did my PhD in Science and Technology Studies. At NYU Tandon these days I teach a variety of classes on the social implications and the social shaping of science, technology and medicine. For example, I teach introductory classes like Science, Technology and Society, as well as upper-level seminars in Critical Global Health, the History of Cyborgs and Cybernetics, and STS Perspectives on Medicine and the Pharmaceutical Industry. My first major research project looked at food allergy activism in the United States. How do patients for example, engage with scientific knowledge and expertise around key interventions, like epinephrine auto injectors? How do doctors and patients interact in clinical settings? And how does this shape how patients understand themselves, their health and the need for activism. And finally, in the course of my research, I went on a number of really fun trips, including two factories that were producing food allergy friendly food, and a number of patient and activist organized conferences and events that highlighted some of the really ingenious kind of solutions and fixes to the problem of food allergies here is just part of a goodie bag for one of those events. In more recent years, I've turned to studying digital technologies, I've done a little bit of work around virtual reality. So here I am in a headset, culminating in a number of forthcoming publications, as well as a white paper with the anti-defamation league about some of the emerging issues around harassment in virtual spaces. And finally, most recently, I'm working on two new projects. Number one, I'm working on a book about feminist cyborg theory for technical non-specialists and student audiences with my colleague Laura Forlano, and also launching a new project looking at the ways that digital health technologies shape reinforces, reproduce ideas about gender and race. So now I want to get into the topic of the day, who is AI for? We're going to explore this question through a couple of examples drawn from some of my areas of research. First, we're going to look at work. And we'll talk a little bit about uses of AI in recruiting, some of the promises and some of the observed outcomes of these systems. Then we'll talk about health care. And we'll talk about AI used for benefits allocation, as well as AI used to shape who has access to health care. And finally, we'll end up with a discussion of several questions that one might want to include as a policymaker in AI systems evaluation, consultation and implementation processes, as well as a few suggestions for additional readings and organizations to follow or be in touch with around some of these questions. And one of the takeaways from my areas of research, healthcare work, thinking about gender and race, is that AI driven systems do not always help the people most in need of support and access to services, even though that's typically what they're intended to do.

5:08 And in fact, sometimes AI driven systems further restrict access to necessary goods and services for people who are already under served. And so, the question to really think about in a policy setting is how can you, how can policymakers make AI driven systems that truly address the needs of the citizenry? So again, we'll look at a couple of examples and then I'll pull out a couple of reflections, a couple of abstractions, questions that you might ask yourself in this process. So, turning now to our first domain of work. So again, we're going to focus here on one particular application of artificial intelligence in the very wide-ranging field of AI for work and labor. And so very specifically, we'll be looking at AI in recruiting. And so, using AI in the recruiting process has a number of promises that sound really great. AI in recruiting promises to do things like correct for human biases by having fewer humans, who might be biased humans, in the loop of the recruiting process. It also promises to evaluate applicants based on proven indicators of success, what has succeeded in a workplace in the past, what are some of those indicators, some of those skills and experiences. It promises to save time and money in recruiting. And finally, a fourth big promise is that it promises to protect the safety and well-being of current employees and clients again, through a rigorous and supposedly unbiased vetting process. But let's look at some of the examples here. So, one question that immediately springs up when you look at how some of these systems have worked out is are these systems looking for indicators of success? Or are they reinforcing simple forms of bias like sexism? So, for example, in 2018, Reuters reported on an Amazon system in which computer models were trained to vet applicants by observing patterns and resumes submitted to the company over a 10-year period. Because the tech industry is predominantly made up of men, especially in technical roles, most of these applications came from men. And so, in effect, this system ended up teaching itself that male candidates were preferable. And consequently, its penalized resumes and included the word women's as in women's chess club, and it downgraded women's downgraded graduates of all women colleges too in particular. The startup Textio, which provides AI driven services to analyze language in various settings, including in recruiting, put out a blog post also in 2018, illustrating how the language of job ads themselves invite certain kinds of applicants. And in this case, they were again, particularly looking at gender. So, you can see some of the breakdown that they shared, of some of the words and phrases that were associated with men versus women in job ads. And you can see how different companies had different balances of some of that language. And in the accompanying blog post Textio wrote, it's common to talk about company culture, as a part of recruiting. Organizations spend significant time and money on shaping the employment brand. But however, you spin it the truth of your cultural environment shows up in the language that your team uses to communicate. Another application of AI in recruiting is its application in platforms that connect people looking for certain kinds of workers directly to those workers' platforms of the type of Uber, although that's not one example that we'll talk about today. One example that was covered extensively, is the company Predictive. This is a service that allows people looking for care workers like babysitters to use the service to scrape those potential employees' social media accounts to look for different indicators of potential success. And so, in this story reported in the Washington Post, the person to pay did hear, Jesse, started looking for a new babysitter for her one-year-old son.

10:04 And she wanted more information than she could get from a standard criminal background check, parent comments and a face-to-face interview. So, she turned to this risk predictive to get a little bit more information. It aimed at scanners at one of the candidates, Facebook, Twitter and Instagram posts. And it offered for Jessica for Jesse, a automated risk rating of a 24 year old woman who is under consideration. It said that she had a very low risk of being a drug abuser but gave her a higher assessment - a two out of five for bullying, harassment, being disrespectful and having a bad attitude. And the system didn't explain why it made that decision, but it undermined Jesse's enthusiasm for hiring this particular young woman. In another piece that came out around the same time by Brian Merchant in Gizmodo, the author actually runs the social media profiles of several of his direct contacts through the predictive system. And he finds some curious results. So, one person, a woman named Kiana, a musician who babysat part time and had never been anything but kind and respectful, according to the author, was flagged as a moderate risk for disrespectful attitude and a low risk for bullying and harassment. By comparison, a another individual, Nick Rutherford, got a close to perfect score. Even though, as the author writes, Nick's Twitter feed was full of vulgar jokes, sexual innuendo, and F bombs. And if you go and look at the article, there's a number of examples of this language. So in this case, there seem to be a mismatch between what an average person would take away from these social media posts and what the system was taking away. And one of the major differences between Kiana and Nick is that Kiana was a black woman and Nick was a white man. One of the unintended consequences of these systems as well is that they invite users to game them. So as I was browsing through and updating myself on some of the technologies out there today, I came across a entire library of videos on YouTube training potential candidates how to train themselves to succeed at an AI analyzed interview video passed through the company Hirevue. And so there's also a question here of what exactly are you measuring? Are you truly measuring what the candidate's potential might be? Or are you measuring someone's social media posts and someone's ability and savviness to prepare for the signals that the AI system is going to be looking for? So again, the reality here is quite different from the promise. While there's a promise that these systems can correct for human biases by having fewer humans in the loop, the biases of the data used to train these tools, such as the data of 10 years of mostly men applicants to Amazon, can perpetuate past and present biases, like racist and sexist assumptions about language use. Rather than evaluating applicants on proven indicators of success, you might end up just hiring more people who are like the ones that are already employed. Rather than saving time, and money and recruiting, it might be worthwhile to ask, even if time is saved, what opportunities to find diverse and innovative talent have been missed? And finally, while the aim is to protect the safety and well-being of potential clients and co-workers, the feature screens, such as a social media post or someone's mannerisms in a recorded video interview, may actually have very little to do with job performance because of how that data is collected or what data the system deems relevant. So who is AI in recruitment for? In these examples, we can see quite clearly who it is not for. In the tech world, it is not for women, and then the world of care work, it is not for black women. The second domain I want to talk about is healthcare. And again, there are a number of promises about what the use of AI in healthcare can deliver. Reductions in benefits fraud, reduction in biased decision making by caseworkers and other care providers, efficiency and scheduling, and automated risk detection that can potentially cap some health indicators the providers might overlook. But again, I want to go through a couple of examples of what this looks like in the real world.

15:09 And first I want to start with the chapter Automating Eligibility in the Heartland from the book Automating Inequality by Virginia Eubanks. This chapter opens with the story of a young girl named Sophie Stipes. At the age of six, Sophie had a number of health problems that required a gastric tube and a special formula to provide nutrition as well as expensive diapers, and the cost of this care could be as high as $6,000 per month. Benefits supporting the cost of her care, however, were automatically withheld in 2008, following the rollout of a joint IBM affiliated computer systems benefits automation service, because her mother failed to sign a new form that she had never received. And in this case, the replacement of human caseworkers who could maybe double check form submissions, try to troubleshoot problems, remind the parent to get things in on time, was replaced by a digital automated system that didn't do any of those things, and ended up producing new vulnerabilities amongst citizens who are already quite vulnerable, like a chronically ill child. And there were some reasons built into the system why this was. Sophie's tragedy wasn't a simple administrative oversight, or a one-time slip up. The system had been calibrated by human designers to default to reducing benefits whenever possible. Whenever there was a mistake benefits would automatically be revoked when the assumption was that a missing data point represented a willful or intentional lack of compliance. And that trigger to reduce or withhold benefits, was meant to respond to that. And Eubanks in her book argues that it while the decision-making apparatus might be newly computerized, these decisions that it made were also directed and shaped by some of the existing incentives of lawmakers and policy makers and some of the internal logics of computer designers themselves. Assumptions that poor people are untrustworthy and sneaky, for example, they are prone to fraudulent claims, and are burdensome on public resources. I want to look at two quick examples of another place where AI is being used in healthcare specifically in regards to healthcare access. The first is scheduling access. So, in a paper by Michele Samorani and Linda Goler Blount and a number of other colleagues, a team of researchers looked at an automated scheduling system that would allocate different times in the schedule and balance patient loads based on historical data that the system had used to learn which patients are most likely to show up to their appointments. So patients who are more likely to show up would be offered times that we're protected and patients who were deemed, according to this machine learning system, less likely to show up or show up on time, were overbooked, so they were put in a slot with more than one patient so, and this was in a study of safety net clinics and the authors write that they revealed how racial bias was woven into the algorithms of electronic health records scheduling systems. Again, overbooking is meant to ensure that providers are fully utilized even if some patients fail to show up for a scheduled appointment. However, if the patients were scheduled in overbooked slots do show up, some of them will experience extra waiting time at the clinic because the provider can only see one patient at a time. So in this case, this modern appointment scheduling system is designed to decide which parents to overbook and which pay parent patients to prioritize through machine learning. And the purpose is to optimize provider time and clinical revenue. However, these algorithms often tended to overbook Black patients in particular. And so the authors write "built into machine learning, apparently, is that for Black patients, timely, quality care can wait". Social scientist Ruha Benjamin also discussed some similar issues, issues related to medical algorithms, automatically downgrading the priority of black patients in her op ed assessing risk automating racism in Science in 2019. The program that she analyzed was a also quite detailed paper done by a team of data scientists looking at how high risk management programs were allocated to different patients. And patients again, were being allocated to this kind of extra oversight of their care through a machine learning based system. And, as Benjamin writes, Obermeyer and colleagues report on one of the first studies to examine the outputs and inputs have an algorithm that predicts health risk, and influences treatment of millions of people.

20:42 They found that because the tool was designed to predict the cost of care as a proxy for health needs, Black patients with the same risk score as white patients tended to be much sicker, because providers spend much less on their care overall. So again, a system that was seeking to be fair to everyone, systematically disadvantaged Black patients, because Black patients were being treated less to begin with. Benjamin goes on to write practically speaking, they're finding means that if two people have the same risk score that indicates that they do not need to be enrolled in a high-risk management program, the health of the Black patients is likely much worse than that of their white counterparts. And then the researchers in this paper that discovered this went beyond the algorithm developed by developers to construct a more fine-grained measure of health outcomes by extracting and cleaning data from health records. And they sought to determine not just the number of conditions or the amount of money spent per patient, but really the underlying severity of their conditions, which they saw as a much more accurate way to determine who needed extra care and oversight. So again, the reality versus the promise is quite different in these examples of AI used in health care. Rather than a reduction of benefits fraud, we actually see erroneous withholding in the Virginia Eubanks case study of benefits due to undetected system errors. Also, rather than a reduction in bias decision making by caseworkers, we actually see a reduction in opportunities for review of system errors and simple mistakes like forgetting to file a single form. Rather than efficiency and scheduling, we see racism in scheduling. And rather than automated risk detection, picking up on indicators providers might miss, we see automatic risk detection that correlates health status to income and to race indicators, perpetuating racism against already underserved Black patients. So again, returning to this question of who is AI in healthcare for? Similar to the employment examples, we see the AI in healthcare is very much not for Black people, and also not necessarily serving the needs of lower income people very well, either. And so I want to present a couple of takeaways or a couple of questions that you might ask yourself, for folks considering or working with AI in public service contexts. And I want to suggest that there are three key points of intervention for reflecting and fixing processes here to hopefully better support the rollout of AI for the good of citizens. Number one is reconsidering motivations for adopting AI based systems. Number two is being careful in reflecting on and analyzing the purpose of the tool under consideration. And number three is the evaluation process. So a few questions for the first, for motivations for adopting AI based systems. It's worthwhile reflecting on are we motivated to adopt AI only to save money, right? If the default is set to remove or restrict benefits, then the effect of the system might be to further disadvantage people who are already disadvantaged or vulnerable. Do we know who has access to necessary public services and who does not pre-AI? As we saw in some of the healthcare examples, understanding who has access before the rollout of an AI system is really necessary for making sure that the AI system isn't perpetuating already existing biases and differentials in access to services and potentially being able to fix those past differences in future systems. Do we have specific benchmarks for improvement, for example, improvement and access to services? Things like the percent of eligible individuals using services employment equity or health equity. Changes in these indicators, indicating more inequity or less access to service might be a trigger to review the actual basis on which an AI driven system is making decisions about access and benefits services. Finally, are there benchmarks for pausing or removing the system? Should some of those benchmarks for improvement not be met? And should the system create new or exacerbating exacerbated inequitable income outcomes?

25:43 Second, the purpose of the tool under consideration. Will the system base future decisions entirely on past outcomes, either for individuals or for the population at large? And again, if outcomes have been unequal in the past, basing future decisions on that past outcome data will perpetuate inequality and injustice. Was a system originally designed to boost revenue or income for for profit entities or to distribute public benefits and services? Right, we see a lot of private sector companies working with different government entities on smart systems and on AI driven decision making systems. Those systems might have been developed for different purposes than the purposes that are often central to public service work. And it's very much worthwhile to ask those hard questions about what the systems were developed for, what their defaults are, what values and qualities they prioritize. Is the system designed to withhold services if in doubt? Is the system designed, for example, to make a family wait for thousands of dollars of benefits needed to keep alive a chronically ill child? And finally, where will humans have oversight in the system decision making and appeals process? If something goes wrong, how can it be fixed? How quickly can it be fixed? And is the speed and efficacy of those fixes properly calibrated to the seriousness of the potential mistakes? And finally, the evaluation process. Do evaluators of these systems have a variety of forms of expertise? People from the technology world might have a lot of expertise in technology. But that does not mean that they have expertise in fields like public health, participatory research, communications, labor, citizens' organizations, or state or municipal government operations. Having multiple experts at the table and involved in the decision-making evaluation and rollout process processes is really important because these systems can have wide ranging unintended consequences that go far beyond the scope of expertise of the technology experts who built them. Does the evaluation process include opportunities for public consultation and redesign of systems and implementation plans, right? And are those opportunities meaningful? Right? Not just listening to citizens, but letting citizens make demands on public service agencies and on the companies designing these systems in order to make meaningful changes to the way they work and to the outcomes that they will have on a community. Do you have a clear definition of who counts as a community or the public that might be affected? Do you even know who is potentially going to be affected? Again, do you have those basic stats of existing users and good projections of future users in order to understand the potential risks and the potential benefits to different parts of your community? And finally, can the community say no? Is there a mechanism for community input to stop the implementation of an AI driven system? I think we've seen time and time again, especially here in my home of New York City, that this is not the case. The community evaluation very rarely stops the implementation of new systems. We've seen this especially in law enforcement uses of AI in New York City, that there's very, very little oversight and virtually no ability to stop the rollout of these systems. If the community can't stop the rollout or development of a new system, then community participation is not meaningful, and you need to go back to the drawing board on that. Finally, I want to end with a couple of items for further reading and some organizations to keep an eye out for. First, I would recommend to take a look at Virginia Eubanks' book, Automating Inequality.

29:57 It's an excellent and lively read, not too jargony, full of really great examples of what works and what doesn't, as well as how public policymakers and public servants are actually working through the day-to-day challenges of learning how to use an automated decision-making system and integrating it into government functions. As Eubanks writes in the introduction, we all inhabit a new regime of digital data, but we don't all experience it in the same way. This is a great book for understanding how different people in a community might experience the same data and data driven system. Second, I want to recommend Ruha Benjamin's book Race After Technology for further reading on technology and racial justice. As Benjamin writes, with emerging technologies, we might assume that racial bias will be more scientifically rooted out yet rather than challenging or overcoming the cycles in and of inequity, technical fixes too often reinforce and even deepen the status quo. This is a great look at how that happens, the consequences when that happens, and also the kinds of community-based organizations and community governance structures that can prevent algorithmic systems from perpetuating injustice in a community. This is a great book not only for critique, but also toolkits, models and organizations to look out for that are charting a path forward with data. Finally, I recommend the book Design Justice by Sasha Costanza-Chock for thinking about community led design. How can a community get meaningfully involved in the discussion design and evaluation process of a new automated system? As Costanza-Chock writes, most designers most of the time, do not think of themselves as sexist, racist, homophobic, xenophobic, Islamophobic, ableist or settler-colonialist. However, design justice is not about intentionality. It is about process and outcomes. This book is filled with great examples of processes that work and is especially attentive to how to incorporate existing technical workflows into a really justice focused way of thinking about design of technological systems for the public. And finally, a couple of organizations that I would love to suggest so you keep an eye out for: Black in AI, which is an organization or a group that brings together Black researchers in AI and fosters community conversation through other platforms and conferences like NeurIPS; the Algorithmic Justice League started by MIT Media Lab graduate student Joy Buolamwini brings together a, a very high powered network of Black technology critics and designers. And finally, Data 4 Black Lives, an organization that thinks not only about automated decision-making systems but other forms of data and how they impact Black people and Black communities in the United States. So that's it. That's "AI for Whom?" Again, I'm Danya Glabau at NYU Tandon School of Engineering. Thank you so much for joining me for this talk today. I hope you enjoyed it. And all of the best with thinking about AI systems, thinking about who they really serve and how you can make them serve the public better.